# Streaming Big Data: Lesson Overview

In this lesson, we'll explore the role of message brokers and learn about the design and implementation of streaming architectures to ensure scalable and innovative data solutions.



### What You'll Learn:

**Understanding Message Brokers: From Kafka Fundamentals to Amazon MSK**

Identify the components and service models of message brokers, with a focus on Apache Kafka and Amazon MSK.

This will enable you to design and implement streaming data pipelines.

## Architectures, Guarantees, and Challenges in Streaming Applications

Understand the common architectures, delivery guarantees, and challenges of streaming applications.

This will enable you to make informed decisions regarding streaming architectures, ensuring they are robust and scalable.

### Ready to Dive In?

**Reflect:**
Think of a real-world system that delivers real-time updates — a social media feed, a stock ticker, or a ride-hailing app. What challenges would arise if that data were delivered in large batches instead of as a continuous stream?